

Week 5, Day 1 — presenter notes

Slides: [week-05-day-01.html](#) → **F** fullscreen → arrows. **N** notes. **P** print.

HTTP, JSON, Gin, first SMS API. Stdlib first. No DB. No JWT. No CORS implementation.

Timing (3–3.5 hours)

Clock	Block	Slides
0:00–0:15	Week 4 skeleton check	1–2
0:15–0:30	Outcomes + why REST now	3–4
0:30–0:50	HTTP + JSON concepts	5–6
0:50–1:20	Live <code>net/http</code> hello	7
1:20–2:10	Gin, health, hello, minimal struct (live)	8–13
2:10–2:30	Status, validation light, curl	14–16
2:30–end	Postman lab + PR	17–end

Never skip stdlib HTTP. Never skip a 200 in Postman.

Live demo script

1. Stdlib (throwaway ok)

```
http.HandleFunc("/hello", func(w http.ResponseWriter, r *http.Request) {
    fmt.Fprintln(w, "hello from stdlib")
})
http.ListenAndServe(":8080", nil)
```

2. Gin health in `cmd/server`

```
cd backend
go get github.com/gin-gonic/gin
go mod tidy
```

```
r := gin.Default()
r.GET("/health", func(c *gin.Context) {
```

```
c.JSON(http.StatusOK, gin.H{
    "status": "ok",
    "message": "Backend is running",
})
})
r.Run(":8080")
```

```
curl -i http://localhost:8080/health
```

Lab gate

- [] Gin in `go.mod`
- [] `GET /health` exact JSON
- [] `GET /api/hello` JSON
- [] curl 200 + Postman 200
- [] Pushed / PR started

Homework

PR + Postman screenshot. Stretch: hard-coded `GET /api/students` array.

Typical failures

Symptom	Fix
<code>connection refused</code>	server not running / wrong port
port already in use	kill old process
package gin not found	<code>go get</code> + tidy from <code>backend/</code>
empty body	wrong path or method
JSON keys differ	match <code>status</code> / <code>message</code> exactly

What not to teach today

JWT, Postgres, CORS code, full CRUD, deep structs, middleware. Tease week 6 only.